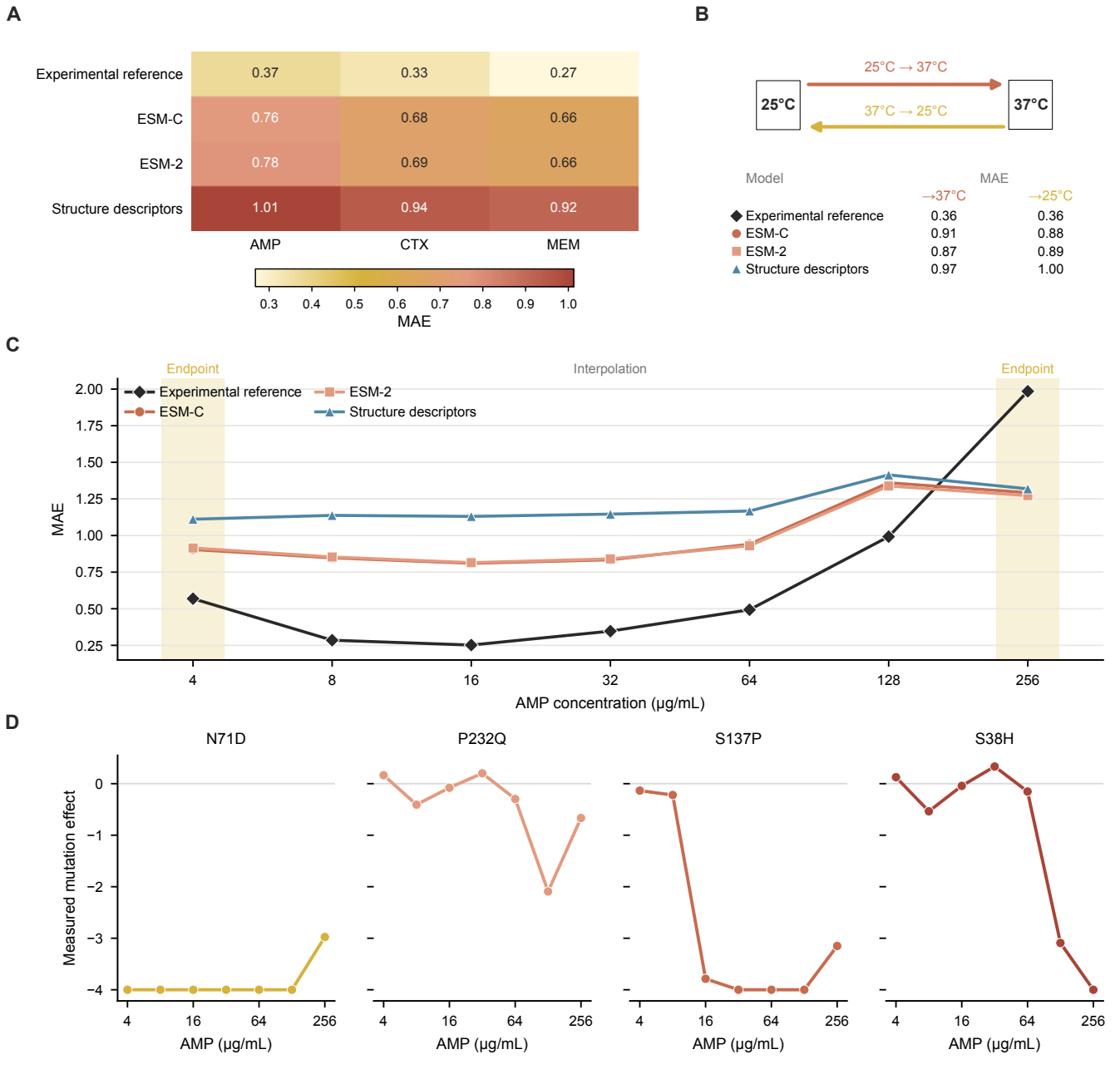

A
Experimental reference
ESM-C
ESM-2
Structure descriptors
0.37
0.33
0.27
0.76
0.68
0.66
0.78
0.69
0.66
1.01
0.94
0.92
AMP
CTX
MEM
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
MAE
B
25°C → 37°C
37°C → 25°C
25°C
37°C
Model
MAE
→37°C
→25°C
Experimental reference 0.36 0.36
ESM-C 0.91 0.88
ESM-2 0.87 0.89
Structure descriptors 0.97 1.00
C
Endpoint
Interpolation
Endpoint
Experimental reference
ESM-2
ESM-C
Structure descriptors
MAE
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
4
8
16
32
64
128
256
AMP concentration (μg/mL)
D
N71D
P232Q
S137P
S38H
Measured mutation effect
0
−1
−2
−3
−4
4
16
64
256
AMP (μg/mL)